Huaman Computer Interaction

Time: 3 Hrs.

Answer All Questions

Question 1

1. Define Human–Computer Interaction (HCI)

   Human–Computer Interaction (HCI) is the study and practice of how people interact with computers. It focuses on designing interactive systems that are usable, efficient, safe, and enjoyable for users. It combines computer science, psychology, ergonomics, and design to improve the communication between humans and machines.

2. What is meant by interaction as communication in HCI?

   *Interaction as communication describes the exchange of information between a **user** and a **system**, similar to a conversation. The user sends input (commands, clicks, gestures), and the system responds with output (visual display, sound, feedback). Effective interaction depends on **clear, meaningful communication** between the two.*

3. What is meant by human perception in the context of HCI?

   Human perception in HCI refers to how users interpret sensory information—primarily vision, hearing, and touch—while interacting with a system. This includes recognizing icons, understanding sound cues, sensing vibrations, and visually interpreting layouts. Designers must consider perceptual limits such as colour blindness, hearing loss, or limited tactile sensitivity.

4. Explain the concept of Human–Computer Interaction and discuss why it is important in modern computing environments.

   *HCI is the discipline that ensures the interaction between humans and digital systems is **intuitive, efficient, and user-friendly**. It is important because:*

   - *Modern technology (applications, websites, devices) is used by millions of diverse users.*
   - *Good HCI reduces errors, increases productivity, and improves satisfaction.*
   - *It lowers training time and support costs.*
   - *It is essential in safety-critical systems such as aviation, healthcare, and banking.*
   - *It enhances usability, accessibility, and adoption of software products.*

(Marks 4*5 = 20)

Question 2

1. Explain the main goals of Human–Computer Interaction (HCI)
    - ***Improve Usability*** *— make systems easy to learn, use, and remember.*
    - ***Enhance User Experience*** *— enjoyable, satisfying interaction.*
    - ***Ensure Efficiency and Effectiveness*** *— complete tasks quickly with minimal errors.*
    - ***Support Accessibility*** *— usable by people with disabilities.*
    - ***Reduce Cognitive Load*** *— avoid overwhelming the user.*

2. Discuss how poor HCI design can increase support and development costs.
   Use real-world examples to justify your answer.
    - ***Increased help-desk calls***
      Example: Confusing form layouts lead to repeated user mistakes.
    - ***More training sessions required***
      Example: Complicated enterprise software requiring long onboarding.
    - ***Frequent redesign and bug fixes*** *because users make errors or misunderstand system behaviour.*
    - ***Higher development costs*** *to repair usability flaws discovered late.*
    - ***Loss of customers***
      Example: A poorly designed e-commerce checkout causes cart abandonment.

   Good usability prevents these issues by reducing errors, simplifying interactions, and supporting predictability.

3. A hospital Intensive Care Unit (ICU) uses a computerized system to monitor patient vital signs and manage critical medical data. However, the system occasionally crashes and sometimes displays incorrect information. Medical staff depend heavily on this system to make quick decisions, and such failures can place patients and staff at serious risk.
    a. Explain two reasons why this issue is critical in an ICU environment.

       Risk to patient safety
       In an ICU, doctors and nurses rely on real-time and accurate vital signs (such as heart rate, blood pressure, and oxygen levels) to make immediate medical decisions. If the system crashes or shows incorrect data, staff may make wrong decisions, which can lead to serious injury or even death of patients.

Delay in emergency response
ICU situations require fast responses. System failures interrupt workflow and force medical staff to spend extra time verifying information or switching to manual methods. These delays can be life-threatening in critical situations where every second is important.

b. Suggest two HCI design improvements to address the problem.

Improve system reliability and error prevention

The system should be designed with strong error prevention mechanisms such as automatic system checks, backup data displays, and fail-safe modes. Clear error messages and warnings should be provided so staff immediately understand when data is unreliable.

Design for critical environment safety

The interface should be simple, clear, and easy to read under pressure, with large fonts, clear color coding, and audible alarms for abnormal values. Backup displays or secondary systems should be available to ensure continuous access to vital information even if the main system fails.

(Marks 5*4 = 20)

Question 3

1. Describe the four stages of the User-Centered Design process. **(Marks 5)**

a. **Specify Context of Use**
Identify users, their environment, tasks, and limitations.

b. **Specify Requirements**
Functional and non-functional needs are documented.

c. **Create Design Solutions**
Develop sketches, wireframes, prototypes, and refine iteratively.

d. **Evaluate Designs**
Usability testing, feedback collection, and design improvements.

2. Evaluate the advantages and disadvantages of the User-Centered Design approach.

(Marks 5)

Advantages:

- Produces usable and effective systems.
- Reduces redesign and development cost.
- Increases user satisfaction and performance.
- Users gain sense of ownership.

Disadvantages:

- Time-consuming process.
- Higher cost due to user involvement.
- Requires skilled designers and researchers.
- Difficult to translate some user data into design

3. You are redesigning an interface for elderly users. Using the human Input–Output channel model (vision, hearing, touch), analyze what sensory limitations must be considered and propose improvements to enhance usability.

(Marks 10)

*When redesigning an interface for elderly users, it is important to consider age-related limitations in the **human input–output channels**, especially **vision, hearing, and touch**. Addressing these limitations improves usability, accessibility, and user satisfaction.*

1. Vision (Visual Channel)

Sensory limitations:

- *Reduced visual acuity makes it difficult to read small text.*
- *Decreased contrast sensitivity causes problems distinguishing similar colors.*
- *Slower visual processing affects understanding of crowded or complex screens.*

Design improvements:

- *Use **larger fonts** with adjustable text size.*
- *Apply **high-contrast color schemes** (e.g., dark text on a light background).*
- *Avoid clutter by using **simple layouts** with sufficient spacing.*
- *Use clear icons with accompanying text labels.*

2. Hearing (Auditory Channel)

Sensory limitations:

- *Age-related hearing loss, especially for high-frequency sounds.*
- *Difficulty understanding alerts in noisy environments.*

Design improvements:

- *Provide **visual alternatives** (pop-up messages, icons) for audio alerts.*
- *Use **lower-frequency, adjustable-volume sounds**.*
- *Allow users to customize alert types (sound, vibration, visual).*

3. Touch (Haptic / Motor Channel)

Sensory limitations:

- *Reduced tactile sensitivity and slower motor responses.*
- *Difficulty performing precise touch actions due to tremors or reduced control.*

Design improvements:

- *Use **large buttons** with enough spacing to prevent accidental touches.*
- *Avoid gestures requiring high precision (e.g., small swipe areas).*
- *Provide **haptic feedback** (vibration) to confirm actions.*
- *Increase response time to avoid time-based errors.*

*By considering limitations in vision, hearing, and touch, interfaces can be redesigned to better suit elderly users. Applying the human input–output channel model helps designers create systems that are **usable, accessible, safe, and user-friendly**, enabling elderly users to interact with technology more confidently and effectively.*

Question 4

1. Explain Abowd and Beale's Interaction Framework. **(Marks 5)**

*The framework consists of **four components**:*

- ***User** – produces input through articulation.*
- ***Input Device** – translates user actions.*
- ***System** – interprets actions in its internal (core) language.*
- ***Output Device** – presents system responses for user observation.*

*Interaction = **translations** between each component's language. Problems occur when translation breaks.*

2. Briefly Explain Donald Norman's Model of Interaction, explaining both the Gulf of Execution and the Gulf of Evaluation.

(Marks 5)

Gulf of Execution
Gap between what a user wants to do and how the system allows it.
Occurs due to confusing controls, hidden options, unclear icons.

Gulf of Evaluation
Gap between system feedback and user understanding of results.
Occurs if feedback is missing, unclear, or delayed.

3. A user attempts to submit an online form but cannot locate the "Submit" button easily. Using Norman's interaction concepts, explain what might be causing the difficulty and how the Gulf of Execution can be reduced.

(Marks 10)

Reason (Gulf of Execution):

- *Button is hard to locate*
- *Poor visibility or placement*
- *Unfamiliar terminology*
- *Bad colour contrast*
- *Inconsistent location compared to other screens*

How to Reduce Gulf of Execution:

- *Make button **visible** (Norman's principle: Make things visible).*
- *Use consistent design and placement.*
- *Add clear labels like "Submit".*
- *Increase button size and contrast.*

Question 5

1. Define Interaction Design

   Interaction Design is the discipline focused on designing ***interactive digital systems*** *that support meaningful communication between users and technology. It involves planning how users perform tasks and how the system responds.*

2. Explain the concept of usability in HCI and discuss how it supports effective system development.
   Usability ensures systems are:
   - *Effective*
   - *Efficient*
   - *Easy to learn*
   - *Easy to remember*
   - *Safe*
   - *Satisfying to use*

   Good usability reduces training costs, increases performance, improves user satisfaction, and reduces errors.

3. State Norman's principle "Make things visible" with an example.

   Principle:
   Important actions and system status should be clearly visible to the user.

   Example:
   A ***visible Submit button*** *or volume slider that clearly shows loudness level.*

4. Write two advantages of using heuristics in HCI.

   - *Detect usability issues early at low cost.*
   - *Do not require real users; evaluations are quick.*
   - *Improve usability, efficiency, and satisfaction.*

5. What is the purpose of error prevention in interface design? Give one example.

Purpose:
To stop users from making mistakes before they occur.

Example:
*Disabling the **Submit** button until all required fields are completed.*

(Marks 5*4 = 20)